Supplementary Material: System Parameters and Input Profiles for Coordinated Dispatch

Danyang Xu, Fengqi Lyu, Jiawei Qu, Zeyu Liu, Hao Wu, Kai Hou, and Hongjie Jia State Key Laboratory of Smart Power Distribution Equipment and System, Tianjin University, Tianjin, China
{danyangxu, lyufengqi, jiawei_qu, tjulzy, hwu, khou, hjjia}@tju.edu.cn

I. Distribution Network Parameters

TABLE I
Common DN Operating Parameters

Parameter	Value	Unit
T; Δt	24; 1	h
Voltage base; power base	12.66; 1	kV; MVA
$\underline{V}_n, \overline{V}_n$	0.9500, 1.05	p.u.
$\overline{I}_{ij}$	200	A
$\overline{P}^0, \overline{Q}^0$	15, 15	MW, Mvar
$\underline{p}_m, \overline{p}_m$	$-1, 1$	MW
c_{loss}	50	CNY/MWh

TABLE II
DN Generator Operating Limits

DG	Bus	$\underline{g}_g^P$	$\overline{g}_g^P$	$\underline{g}_g^Q$	$\overline{g}_g^Q$	$R_g^{\uparrow} = R_g^{\downarrow}$
1	4	0.0000	0.8000	-0.5000	0.8000	0.6000
2	7	0.0000	0.8000	-0.5000	0.8000	0.6000
3	14	0.0000	0.8000	-0.5000	0.8000	0.6000

TABLE III
DN Generator Cost Coefficients

DG	a_g^{DN}	b_g^{DN}	c_g^{DN}
1	25.94	293.18	4.27
2	68.79	406.52	3.42
3	70.89	253.15	4.53

TABLE IV
DN NODAL LOAD ALLOCATION ENTRIES

Bus	P_n	Q_n	Bus	P_n	Q_n
1	0	0	18	90	40
2	100	60	19	90	40
3	90	40	20	90	40
4	120	80	21	90	40
5	60	30	22	90	40
6	60	20	23	90	50
7	200	100	24	420	200
8	200	100	25	420	200
9	60	20	26	60	25
10	60	20	27	60	25
11	45	30	28	60	20
12	60	35	29	120	70
13	60	35	30	200	600
14	120	80	31	150	70
15	60	10	32	210	100
16	60	20	33	60	40
17	60	20	–	–	–

II. MICROGRID PARAMETERS

TABLE V
MG RESOURCE CONFIGURATION AND PEAK LOAD

MG	Bus	$\lvert\mathcal{R}_m\rvert$	$\lvert\mathcal{G}_m\rvert$	$\lvert\mathcal{B}_m\rvert$	$\max_t L_{m,t}$ (MW)
1	2	2	5	4	1.85
2	9	2	3	3	1.40
3	15	2	5	5	1.71

TABLE VI
MG PV NAMEPLATE CAPACITIES

MG	PV unit	$\overline{p}_{m,a}^{\mathrm{PV,rated}}$
1	1	0.2562
1	2	0.1937
2	1	0.2517
2	2	0.2782
3	1	0.1817
3	2	0.2606

TABLE VII
MG GENERATOR OPERATING PARAMETERS

MG	DG unit	$\underline{p}_{m,g}^{G}$	$\overline{p}_{m,g}^{G}$	$R_{m,g}^{\mathrm{MG}}$
1	1	0	0.3482	0.2612
1	2	0	0.4044	0.3033
1	3	0	0.2821	0.2116
1	4	0	0.3356	0.2517
1	5	0	0.3870	0.2902
2	1	0	0.4624	0.3468
2	2	0	0.2803	0.2103
2	3	0	0.3888	0.2916
3	1	0	0.2128	0.1596
3	2	0	0.2618	0.1964
3	3	0	0.3602	0.2701
3	4	0	0.3094	0.2321
3	5	0	0.4268	0.3201

TABLE VIII
MG STORAGE POWER AND ENERGY LIMITS

MG	ESS	$\overline{p}^{\mathrm{ch}}_{m,b} = \overline{p}^{\mathrm{dis}}_{m,b}$	$\underline{e}_{m,b}$	$\overline{e}_{m,b}$	$e^{\mathrm{ini}}_{m,b}$
1	1	0.1209	0.0363	0.3627	0.1813
1	2	0.1832	0.0550	0.5496	0.2748
1	3	0.2722	0.0817	0.8165	0.4083
1	4	0.2955	0.0886	0.8864	0.4432
2	1	0.2302	0.0691	0.6905	0.3453
2	2	0.2009	0.0603	0.6027	0.3013
2	3	0.2426	0.0728	0.7278	0.3639
3	1	0.2160	0.0648	0.6480	0.3240
3	2	0.2223	0.0667	0.6669	0.3335
3	3	0.1677	0.0503	0.5030	0.2515
3	4	0.1043	0.0313	0.3130	0.1565
3	5	0.1240	0.0372	0.3721	0.1860

TABLE IX
MG STORAGE CHARGING AND DISCHARGING EFFICIENCIES

MG	ESS	$\eta^{\mathrm{ch}}_{m,b}$	$\eta^{\mathrm{dis}}_{m,b}$
1	1	0.8676	0.8909
1	2	0.9120	0.9191
1	3	0.8753	0.9466
1	4	0.9494	0.9174
2	1	0.9238	0.8981
2	2	0.8639	0.8593
2	3	0.8928	0.8856
3	1	0.8679	0.8590
3	2	0.8696	0.8825
3	3	0.9073	0.9093
3	4	0.9389	0.9021
3	5	0.9290	0.8627

TABLE X
COMMON MG OPERATING-COST COEFFICIENTS

Symbol	Value	Unit
$a^{\mathrm{MG}}_{m,g}$	2	CNY/(MW^{2}h)
$b^{\mathrm{MG}}_{m,g}$	600	CNY/MWh
$c^{\mathrm{ch}}_{m,b}$	100	CNY/MWh
$c^{\mathrm{dis}}_{m,b}$	100	CNY/MWh
$c^{\mathrm{curt}}_{m,a}$	50	CNY/MWh

III. DAILY LOAD AND PV PROFILES

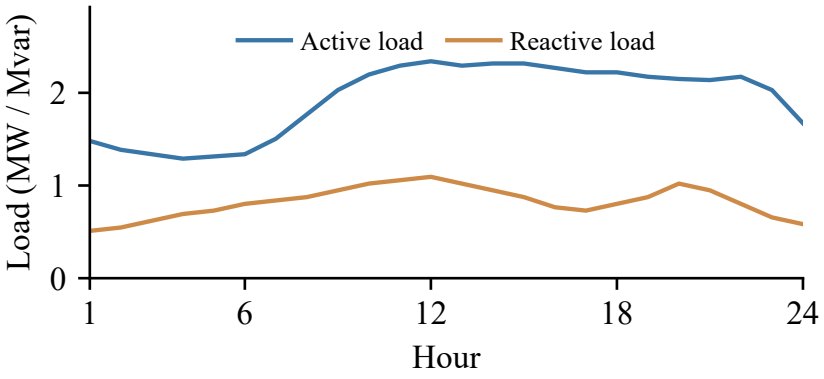


Fig. 1. Daily DN background active and reactive loads.

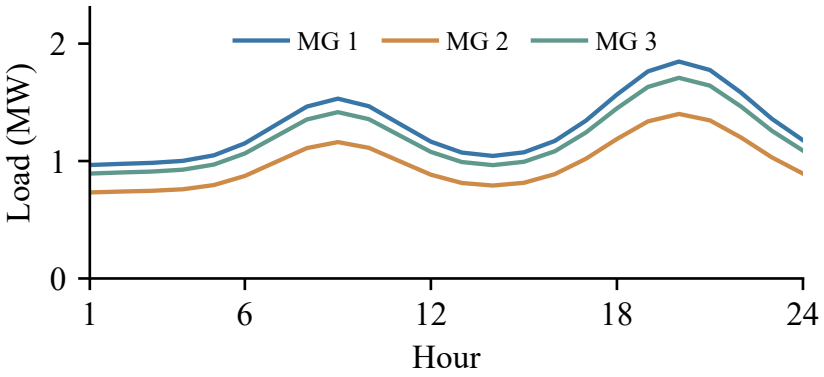


Fig. 2. Daily load profiles of the three MGs.

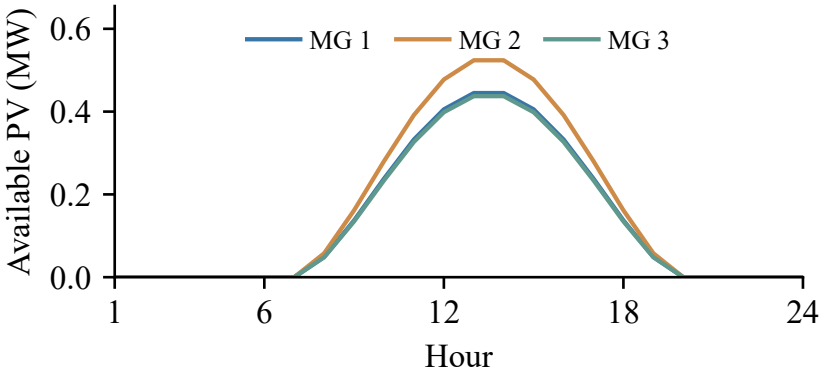


Fig. 3. Daily available PV profiles of the three MGs.

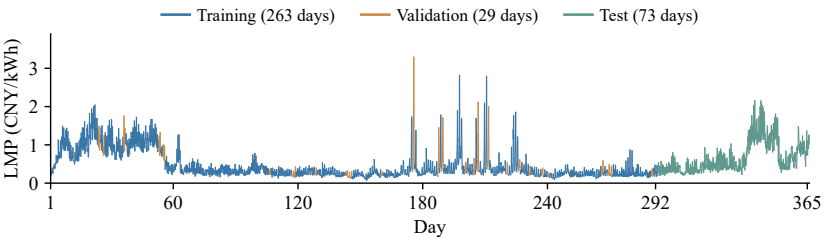


Fig. 4. Annual LMP profiles at a fixed exchange rate of 7 CNY/USD.